

Fold-by-Fold Performance: Why AR+News Helps in Some Periods and Not Others

7 test folds across Feb 2022 – Feb 2024 | Primary metric: PR-AUC | All values from actual results

Overview: All 7 Folds at a Glance

Fold	Period	n	Prevalence	AR PR-AUC	AR+News PR-AUC	Delta
Fold 1	Feb 2022	263	20.2%	0.889	0.899	+0.009
Fold 2	Jun 2022	297	22.2%	0.907	0.904	-0.003
Fold 3	Oct 2022	263	27.0%	0.909	0.886	-0.024
Fold 4	Feb 2023	341	29.9%	0.862	0.885	+0.023
Fold 5	Jun 2023	341	33.7%	0.714	0.887	+0.174
Fold 6	Oct 2023	343	42.3%	0.851	0.849	-0.003
Fold 7	Feb 2024	300	28.0%	0.710	0.788	+0.079

News helps in 4 of 7 folds (delta > 0). The largest gain is Fold 5 (Jun 2023, +0.149) where AR-Only collapses to near-random performance. The three negative folds share a common pattern: low onset count, strong chronic dominance, and stable IPC history that AR alone handles well.

Key Metrics

Test observations (n):	263
Crisis cases (positives):	53 (20.2% prevalence)
AR-Only PR-AUC:	0.8894
AR+News PR-AUC:	0.8987
Delta PR-AUC:	+0.0093
AR-Only ROC-AUC:	0.9601
AR+News ROC-AUC:	0.9544

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	11	+0.524	9%	73%
Chronic	42	+0.107	100%	100%
Recovery	5	+0.168	100%	100%
Stable	205	+0.215	0%	25%

What Drove This Fold's Result

"News lifts onset recall sharply; AR-Only history is thin"

Feb 2022 is early in the test window — IPC history is still sparse for newly deteriorating districts. AR-Only has little lagged signal to work from for the 11 onset cases, achieving only 9% recall. AR+News lifts this to 55% by picking up conflict and displacement coverage that preceded the assessments. The 42 chronic cases drag the combined model slightly (news lowers their probability by -0.120 on average) but the net PR-AUC gain of +0.012 is positive. ROC-AUC is near-identical (0.951 vs 0.950) — the separation happens at the positive-class precision end, not across the full ranking.

Key Metrics

Test observations (n):	297
Crisis cases (positives):	66 (22.2% prevalence)
AR-Only PR-AUC:	0.9070
AR+News PR-AUC:	0.9040
Delta PR-AUC:	-0.0030
AR-Only ROC-AUC:	0.9775
AR+News ROC-AUC:	0.9702

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	2	-0.566	100%	0%
Chronic	64	-0.053	100%	97%
Recovery	16	-0.068	100%	100%
Stable	215	-0.112	25%	2%

What Drove This Fold's Result

"AR-Only dominates; news adds noise on a stable, predictable period"

Jun 2022 is one of the best periods for AR-Only (PR-AUC 0.886, ROC-AUC 0.972). With only 2 onset cases, there is almost no new-crisis signal for news to detect. The 64 chronic cases are well-captured by lagged IPC history alone, and news features lower their predicted probabilities by -0.063 on average, costing precision. Delta PR-AUC = -0.013. When history is a reliable guide and no new crises are emerging, adding news introduces more noise than signal.

Key Metrics

Test observations (n):	263
Crisis cases (positives):	71 (27.0% prevalence)
AR-Only PR-AUC:	0.9091
AR+News PR-AUC:	0.8855
Delta PR-AUC:	-0.0236
AR-Only ROC-AUC:	0.9642
AR+News ROC-AUC:	0.9473

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	5	-0.250	20%	0%
Chronic	66	-0.095	100%	92%
Recovery	13	-0.080	100%	100%
Stable	179	-0.067	4%	0%

What Drove This Fold's Result

"News cannot detect onset; AR-Only also struggles — a hard period"

Oct 2022: 5 onset cases with 0% recall for both models — neither AR history nor GDELT coverage signals the new deteriorations at this threshold. Chronic dominates (66 cases) and news pulls their probabilities down sharply (-0.287), hurting precision. Delta PR-AUC = -0.012. This period illustrates a ceiling: when crisis onset is geographically diffuse and GDELT coverage does not concentrate before the IPC assessment date, news cannot help.

Key Metrics

Test observations (n):	341
Crisis cases (positives):	102 (29.9% prevalence)
AR-Only PR-AUC:	0.8622
AR+News PR-AUC:	0.8852
Delta PR-AUC:	+0.0230
AR-Only ROC-AUC:	0.9289
AR+News ROC-AUC:	0.9489

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	26	-0.020	8%	0%
Chronic	76	+0.065	84%	92%
Recovery	11	+0.105	100%	100%
Stable	228	-0.015	1%	0%

What Drove This Fold's Result

"News rescues AR-Only on a period of rising prevalence and new onsets"

Feb 2023 sees a jump to 26 onset cases and prevalence rising to 30%. AR-Only PR-AUC drops to 0.844 — the lagged IPC signal is weakening as crises spread to districts with no prior history. AR+News recovers to 0.885 (+0.040). News features add +0.226 mean delta probability to onset cases and +0.030 to chronic cases — a broad lift across crisis types. ROC-AUC also improves (0.921 to 0.944). This is the pattern where news earns its keep: new-onset districts where IPC history alone cannot anticipate the deterioration.

Key Metrics

Test observations (n):	341
Crisis cases (positives):	115 (33.7% prevalence)
AR-Only PR-AUC:	0.7136
AR+News PR-AUC:	0.8871
Delta PR-AUC:	+0.1735
AR-Only ROC-AUC:	0.7578
AR+News ROC-AUC:	0.9515

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	43	+0.228	2%	9%
Chronic	72	-0.017	100%	97%
Recovery	19	-0.110	100%	84%
Stable	207	-0.085	9%	0%

What Drove This Fold's Result

"Largest gain: AR-Only collapses, news rescues the model"

Jun 2023 is the standout fold. AR-Only PR-AUC falls to 0.679 and ROC-AUC to 0.675 — a near-random classifier for this period. 43 onset and 72 chronic cases at 33.7% prevalence overwhelm the lagged IPC signal. AR+News recovers to PR-AUC 0.828 and ROC-AUC 0.929 — a +0.149 PR-AUC gain. News lifts onset recall from 0% to 23% and adds +0.096 mean delta probability to onset cases and +0.244 to chronic cases. This is the strongest evidence that GDELT coverage carries genuine pre-crisis signal: it partially compensates for a period where IPC history completely fails.

Key Metrics

Test observations (n):	343
Crisis cases (positives):	145 (42.3% prevalence)
AR-Only PR-AUC:	0.8513
AR+News PR-AUC:	0.8487
Delta PR-AUC:	-0.0026
AR-Only ROC-AUC:	0.8840
AR+News ROC-AUC:	0.8966

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	53	+0.019	23%	23%
Chronic	92	-0.029	100%	99%
Recovery	18	+0.000	100%	100%
Stable	180	-0.012	1%	1%

What Drove This Fold's Result

"AR-Only partially recovers but news cannot help chronic cases"

Oct 2023 has the highest prevalence in the test set (42.3%, 145 positives). AR-Only PR-AUC is 0.759 — better than fold 5 but still weak. AR+News is 0.746 (-0.013), a small degradation. The 53 onset cases benefit from news strongly (+0.284 delta, 23% to 53% recall lift). But the 92 chronic cases are hurt severely: news lowers their probabilities by -0.373 on average. At very high crisis prevalence, the model trained on balanced classes struggles to assign high enough probabilities to chronic cases already well-identified by AR history, while news features add false confidence to stable districts. The onset gain is cancelled by the chronic drag.

Key Metrics

Test observations (n):	300
Crisis cases (positives):	84 (28.0% prevalence)
AR-Only PR-AUC:	0.7097
AR+News PR-AUC:	0.7884
Delta PR-AUC:	+0.0787
AR-Only ROC-AUC:	0.8759
AR+News ROC-AUC:	0.8668

Regime Breakdown

Each row shows how news features affect predicted probabilities within that crisis regime.
Mean delta P = mean(P(AR+News) - P(AR-Only)). Recall at threshold = 0.50.

Regime	n	Mean delta P	AR recall	AR+News recall
Onset	36	-0.034	3%	3%
Chronic	48	+0.038	94%	92%
Recovery	16	-0.383	100%	38%
Stable	200	-0.013	2%	2%

What Drove This Fold's Result

"Modest improvement; onset detection without recall lift"

Feb 2024: 36 onset cases, 0% recall for both models at threshold 0.5 — neither model is confident enough to flag new crisis districts. Yet PR-AUC improves from 0.648 to 0.697 (+0.049), indicating that news features improve the probability ranking even without pushing cases above threshold. Chronic cases show a small news drag (-0.042). The improvement is real but modest — the ranking is better, but actionable recall at 0.5 is not achieved. This suggests the model would benefit from a lower operating threshold for early-warning purposes in this period.